

Program : Diploma in Tool & Die Engineering	
Course Code : 6118	Course Title: Jigs & Fixture Lab
Semester : 6	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide the concepts and use of the jigs and fixtures
- To impart professional competency in the manufacturing techniques of jigs and fixtures.
- To introduce the practical application of limit, fit, tolerances & surface roughness.
- To explain complicated shopfloor drawings, produce jobs, and inspect the job for specified dimensions.

Course Prerequisites:

Topic	Course code	Course name	Semester
Introduction to Jigs & fixtures		Production Engineering II	3
Basic design elements of Jigs & fixtures		Tool Engineering Drawing	3
Detailed design aspects of jigs & fixtures		Tool Design	5

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Produce the basic components of Jigs and fixtures	9	Applying
CO2	Apply the acquired knowledge to manufacture simple jigs	15	Applying
CO3	Produce simple fixtures	12	Applying
CO4	Interpret the concepts of Limits, Fit, Tolerance, surface roughness values and symbols.	6	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						3
CO3	3						
CO4	3						3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Produce the basic components of Jigs and fixture.		
M1.01	Illustrate basic locating elements – cylindrical, conical, diamond pin locators, “V” locators etc.	3	Applying
M1.02	Illustrate locating principles – six-point, redundant, fool proofing etc.	2	Applying
M1.03	Illustrate clamping methods – Latch, screw, cam etc.	2	Applying
M1.04	Illustrate the different types of guide bushing	2	Applying
CO2	Apply the acquired knowledge to manufacture simple jigs		
M2.01	Making of simple liner bushes – headed & headless/press fit bushing/ slip & renewable bushing.	6	Applying
M2.02	Making of simple template/latch/box/angle/sandwich/trunnion Jigs	9	Applying
	Lab Exam – I	1.5	
CO3	Produce simple fixtures		
M3.01	Produce simple milling/turning fixture	12	Applying
CO4	Interpret the concepts of Limits, Fit, Tolerance, surface roughness values and symbols.		
M4.01	Apply the Shaft & Hole dimensions for various types of fits using the standard tolerance chart.	2	Applying
M4.02	Demonstrate the types of fits used in press tools, Moulds, jigs & fixtures and calculation of its dimensions for the corresponding basic size.	2	Applying

M4.03	Apply the various surface finish values and symbol used in press tool & mould machining operations for attaining respective surface finish.	2	Applying
	Lab Exam – II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	Jigs & Fixtures P H Joshy
R1	Fundamentals of Tool Design, 6th Edition, 2010, SME Publication
R2	Tool Design, Cyril Donaldson, Special Indian Edition, 2012, Tata Mc Graw Hill.
R3	A Text Book of Production Engineering, P.C. Sharma, 2013, S. Chand& Co

Online Resources:

Sl.No	Website Link
1	https://onlinelibrary.wiley.com/doi/10.1002/9781119191414.oth1
2	https://www.ijser.org/researchpaper/Jig-and-Fixture-Design.pdf
3	http://www.nitc.ac.in/dept/me/jagadeesha/mev303/CHAPT_INTRODUCTION_TO_JIGS_AND%20FIXTURES.pdf